

PAUL OFFEI

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RESEARCH INTERESTS

My research interest is in Machine Learning for natural language & visual understanding and generation with emphasis on Multimodal & Multilingual LLM Safety and Alignment. My focus is developing algorithmic solution or approach to address the risk posed by Multilingual Large Language Models (MLLM) by highlighting open-ended questions on Value Misalignment, Robustness to Attack, Misuse, Autonomous AI Risks, Agent Safety, Interpretability for MLLM Safety, Strategies to MLLM Safety in Practice, AI Governance and Policy for MLLM Safety.

EDUCATION

IU International University of Applied Sciences BSc in Applied Artificial Intelligence	Erfurt, Germany Feb, 2026-date
University of Ghana MSc in Computer Science	Accra, Ghana Jun,2023-Aug,2024
University of Ghana BA in Economics	Accra, Ghana Sep,2015-May,2019

EXPERIENCE

ResolutionAI Pre-doctoral Researcher	Accra, Ghana May-Aug,2025
• Synthesized papers around LLMs and AI Agents using prompt engineering for filtering to classify them into two categories; relevant and not relevant based on their title and abstract and manually evaluated 30% of the papers with 99% accuracy, with all errors being false negatives using open source models. • Performed experiment and reverse engineering of LLM360 models to understand the underlying mechanisms of LLMs.	
LG AI Research, collabrations with DataMaker Research Assistant	Accra, Ghana Aug,2023-Aug,2024
• Involved in bilingual alignment multi-turn datasets curation for supervised fine-tuning and direct preference optimization to improve EXAONE 3.0 instruction following ability and align to human preferences and ethics. • Collaborated with team members to craft question-answer datasets with its corresponding "Pass" or "Fail" annotation, designed to cover a wide range of unethical and insecure scenarios including hate, bias, illegal content, information hazards such as the use of private data in training for the evaluation of EXAONE 3.0 7.8B instruction-tuned model where the model	

produces appropriate responses 84% of the time when faced with carefully curated adversarial queries.

TexkLab

Machine Learning Engineer

Cape Coast, Ghana

Jan,2021-Dec,2023

- Spearheaded the end-to-end development of an AI-powered agritech solution, "Real Time Automatic Detection of Mosaic Virus in Crops," specifically designed to mitigate crop loss across high-value horticultural sectors including tomatoes, peppers, and beans.
- Architected the project's technical roadmap by selecting and implementing a Convolutional Neural Network (CNN) architecture, leveraging Microsoft Azure GPU cloud infrastructure for high-performance model training.
- Conducted rigorous real-time field testing using OAK-D spatial AI cameras to ensure the product met user-centered requirements for speed and accuracy in diverse environmental conditions.

Ministry of Local Government and Rural Development

Data Analyst

Accra, Ghana

Aug,2019-Dec,2020

- Developed Excel-based financial models that improved the accuracy of revenue and expenditure predictions by 25%, reducing the need for mid-year supplementary budget adjustments.
- Involved in the drafting and implementation of the Fee-Fixing Resolution, resulting in an IGF performance that exceeded the annual budget by over 3% through optimized rate and license collection.
- Conducted in-depth variance analysis for major infrastructure projects, identifying 15% in potential cost savings by reallocating underutilized funds to high-priority sanitation and education initiatives.